Chapter 4 — Work, Energy and Power

Q.1 — MULTIPLE CHOICE QUESTIONS (8 × 1 = 8)

1. Work by variable force is: (a) area under F–d graph (b) F/d (c) F + d (d) slope of F–d graph

2. Which is conservative? (a) gravity (b) viscosity (c) friction (d) air drag

3. Power equals: (a) F × d (b) E × t (c) W/t (d) W × t

4. 1 kWh equals: (a) 3600 J (b) 3.6×10^6 J (c) 3.6×10^3 J (d) 1000 J

5. Absolute gravitational P.E.: (a) GMm/R (b) $\frac{1}{2}mv^2$ (c) −GMm/R (d) mgh

6. Escape velocity: (a) 2gR (b) gR (c) √(gR) (d) √(2gR)

7. Escape velocity from Earth: (a) 1.0 km/s (b) 7.9 km/s (c) 11.2 km/s (d) 3.0 km/s

8. With friction, mechanical energy: (a) is created (b) is destroyed (c) changes to heat (d) is unchanged

Q.2 — SHORT QUESTIONS (6 × 2 = 12)

1. Why is the work done by a conservative force zero over a closed path?

2. Why is friction called a non-conservative force?

3. Why is the kilowatt-hour a unit of energy and not power?

4. Why can mgh not be used for a rocket leaving the Earth?

5. Why is gravitational potential energy negative?

6. Why is escape velocity independent of the mass of the body?

Q.3 — LONG QUESTIONS (2 × 8 = 16)

Q.3 (a) Derive the expression for absolute gravitational potential energy. **(4)**

(b) Use it to derive escape velocity and find its value near the Earth's surface. **(4)**

Q.4 (a) Differentiate between conservative and non-conservative forces with examples. **(4)**

(b) Explain the interconversion of kinetic and potential energy and the effect of friction. **(4)**

studysapiens.com